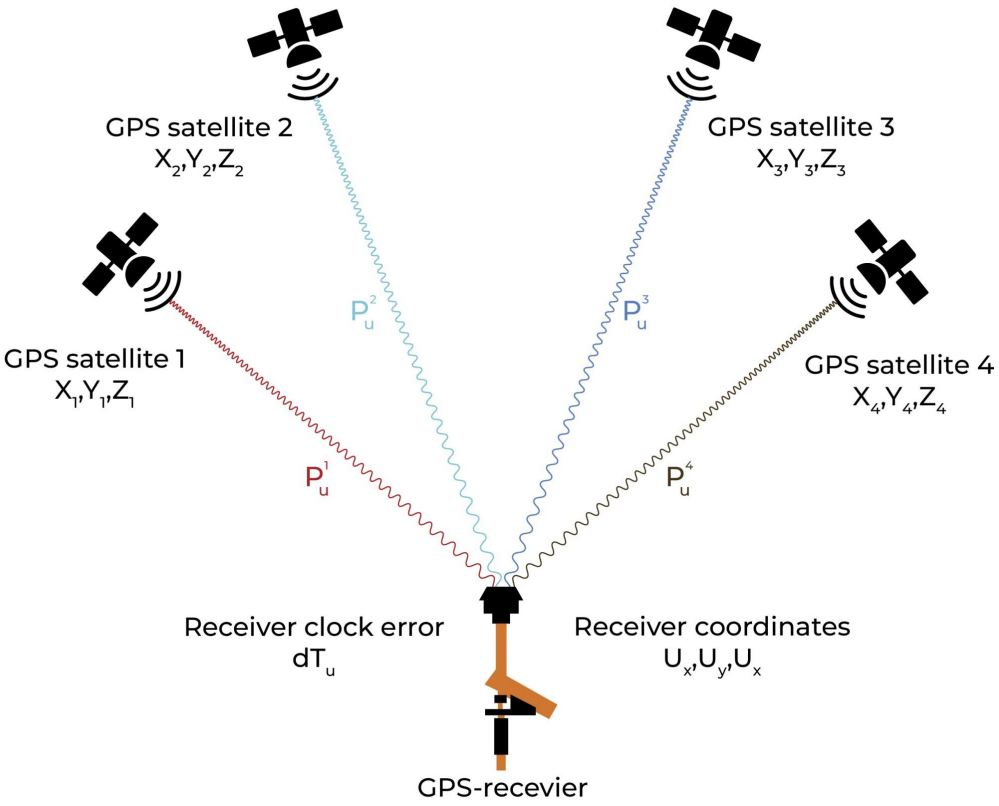
# Assignment 3: GNSS Positioning Models and Differential Techniques

**CE334 Modern Methods in Geoinformatics** 2025-26 II
Instructor: Balaji Devaraju [dbalaji] TA: Kamoosh[kamooshsb24], Ashis[ashisa24]

Issued: 09.02.2026 Submission: 13.02.2026 10:00 h IST

1. **Navigation Solution :** One GPS receiver observing four satellites $S_1, S_2, S_3, S_4$.

   Starting from the geometric range equation, derive the complete pseudorange observation model for receiver in Earth-Centered Earth-Fixed (ECEF) coordinates. Show why a minimum of four satellites are required to determine receiver position and clock bias.




**Hint:**

Use the geometric range equation

$$\rho_i = \sqrt{(U_x - X_i)^2 + (U_y - Y_i)^2 + (U_z - Z_i)^2}$$

and the pseudorange model

$$P_i = \rho_i + c\,dt_i$$

Identify the unknown parameters and express the observation equations in matrix form.

2. **Single and Double Difference Equations :** Two GNSS receivers **q** and **r** simultaneously observe two satellites **a** and **b** using carrier phase measurements.

   Starting from the fundamental carrier phase observation equation, derive the single-difference and double-difference observation equations. Clearly show mathematically which error terms are eliminated at each differencing step and express the final double-difference model in terms of baseline components and integer ambiguities.

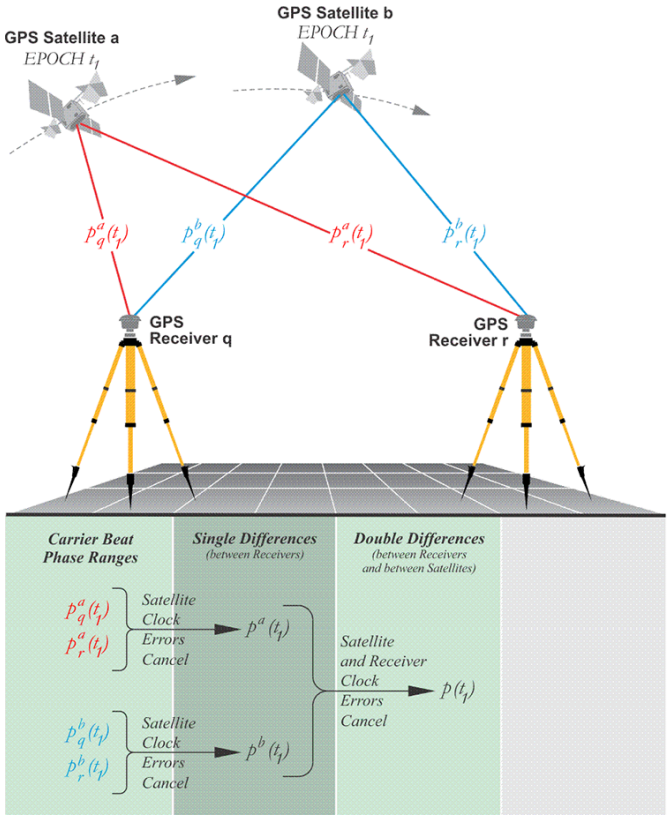
   **Reference:** Hofmann-Wellenhof, B., Lichtenegger, H., & Wasle, E. (2008). *GNSS – Global Navigation Satellite Systems*. Springer, Chapter 6.

   **Hint:**

$$\Phi_r^s = \rho_r^s + c(\delta t_r - \delta t^s) + \text{Troposphere}_r^s - \text{Ionosphere}_r^s + \lambda N_r^s + \varepsilon_r^s$$

   where,

| | |
|---|---|
| $s$ = satellite | $r$ = receiver |
| $\rho_r^s$ = geometric range | $c$ = speed of light |
| $\delta t_r$ = receiver clock error | $\delta t^s$ = satellite clock error |
| $\lambda$ = carrier wavelength | $N_r^s$ = integer ambiguity |
| $\varepsilon_r^s$ = noise + multipath | |




3. **RTK vs DGPS :** Explain, with appropriate mathematical reasoning and observation models, the fundamental differences between RTK and DGPS in terms of measurement type, error mitigation strategy, achievable accuracy, ambiguity handling and baseline length & etc...

# GPS Positioning Using Least Squares Bonus (+25%)

A GNSS receiver observes six satellites simultaneously. The satellite coordinates in the ECEF reference frame (meters) and the observed pseudoranges are given below.

## Satellite Coordinates (ECEF)

| Satellite | $X_i$ (m) | $Y_i$ (m) | $Z_i$ (m) |
|---|---|---|---|
| S1 | 15600000 | 7540000 | 20140000 |
| S2 | 18760000 | 2750000 | 18610000 |
| S3 | 17610000 | 14630000 | 13480000 |
| S4 | 19170000 | 6102000 | 18390000 |
| S5 | 17800000 | 10200000 | 17600000 |
| S6 | 16500000 | 12300000 | 21500000 |

## Observed Pseudoranges

$$P_1 = 21456789.123 \text{ m}$$

$$P_2 = 21123456.456 \text{ m}$$

$$P_3 = 22987654.789 \text{ m}$$

$$P_4 = 22456789.321 \text{ m}$$

$$P_5 = 21934567.654 \text{ m}$$

$$P_6 = 21765432.987 \text{ m}$$

## Approximate Receiver Coordinates

$$X_0 = 1{,}000{,}000 \text{ m}$$

$$Y_0 = 2{,}000{,}000 \text{ m}$$

$$Z_0 = 3{,}000{,}000 \text{ m}$$

$$cdt_0 = 0$$

## Question

Starting from the nonlinear pseudorange observation equation,

$$P_i = \sqrt{(X - X_i)^2 + (Y - Y_i)^2 + (Z - Z_i)^2} + cdt_r$$

linearize the observation model using first-order Taylor expansion, construct the design matrix and perform one iteration of the least squares adjustment to estimate the corrections to $X, Y, Z$ and the receiver clock bias. Also compute the degrees of freedom and comment on the redundancy of the system (The numerical computations should be performed using code, and the obtained results must be included in the submitted report. Students must present the observation equations, linearization procedure, and the matrix formulation (A,X,L,V) along with the solution steps.).

**Note :** Submit handwritten copy.